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Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo

Department of Electrical, Electronic and Computer Engineering

ESP411

DSP Programming and Application

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Assistant lecturers

Bernard Nortier	esp411als@gmail.com	ENG1 13-20
Nicolai Kuschke	esp411als@gmail.com	CFIM 2-18

Assistant lecturer availability

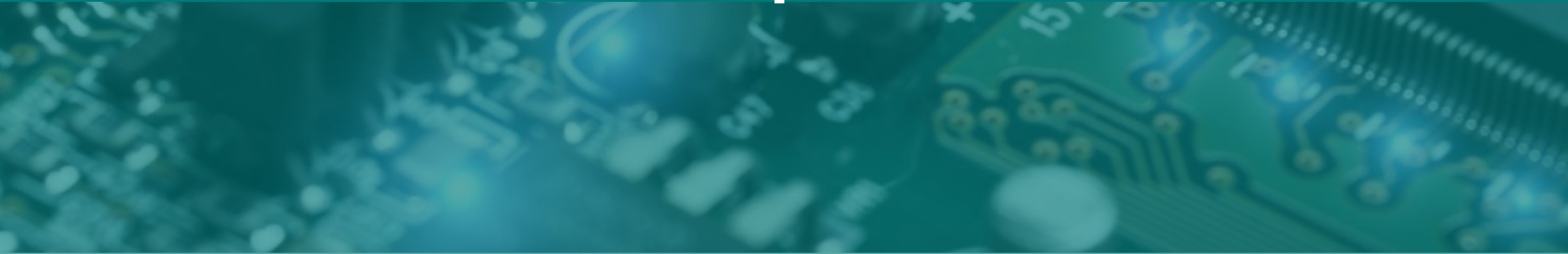
Open lab sessions: Every Tuesday 10:30-13:30

Consultations: By appointment

Email: Ask questions directly, schedule a consultation or request we come by the labs.

Telegram: We will occasionally answer questions on the group but NO DMs.

Pre-Practical Lecture 1: Implementation of an FFT on the STM Development Board



AIM

- The first practical is the implementation of a spectrum analyser for audio and requires:
  - Sampling the audio signal using the STM32F429 triple interleaved ADC
  - The implementation of an FFT algorithm from first principles on the STM32F429 Discovery board
  - The application of the FFT to an input signal (single sinusoid and audio)
  - The display of the output on the screen of the STM board

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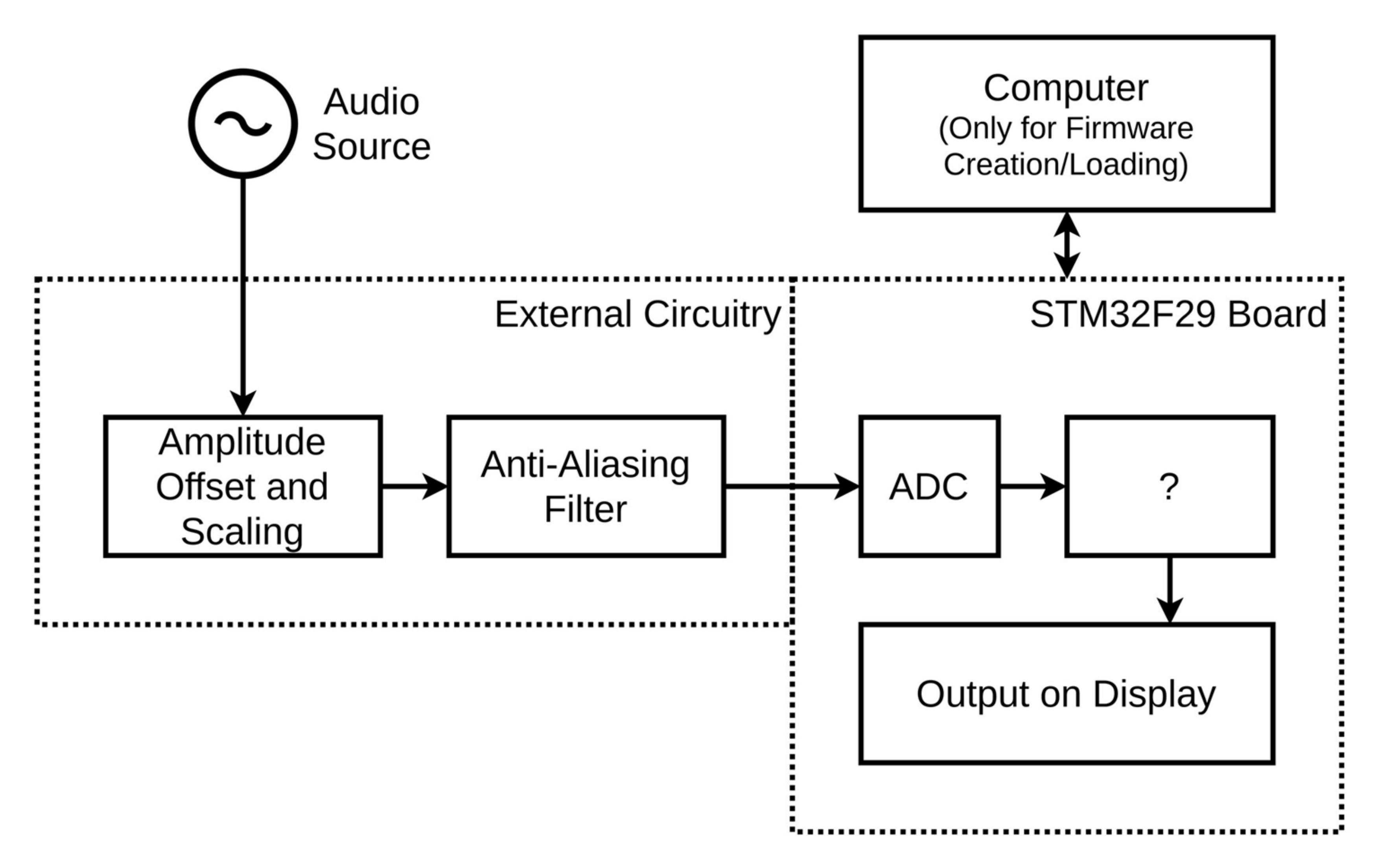
The Required System



Specifications

- Audio source/Signal generator
- -0.5 to 0.5 V input signal (1Vpp)
- 10 Hz to 20 kHz
- Anti-aliasing filter and scaling circuit (external to the board)
- FFT with a minimum of 128 points (may use more) from first principles.

  On-screen display with at least 1 pixel per FFT bin.
- On-screen graph with axes labels (with units in dB (vertical) and kHz (horizontal) scale)

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Fourier Transform

- Method of converting a function of time or space into a function of frequency
- The Fourier Transform cannot be practically implemented
- The Fourier Transform equation is given by:

$$X(f) = \int_{-\infty}^{\infty} x(t) \times e^{-i2\pi ft}\, dt$$

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Discrete Fourier Transform

- DFT: The application of a Fourier transform on a discrete signal
- FFT: An algorithm to implement a DFT
- The DFT equation is given by:

$$X(m) = \sum_{n=0}^{N-1} x(n) e^{-\frac{j 2 \pi m n}{N}}$$

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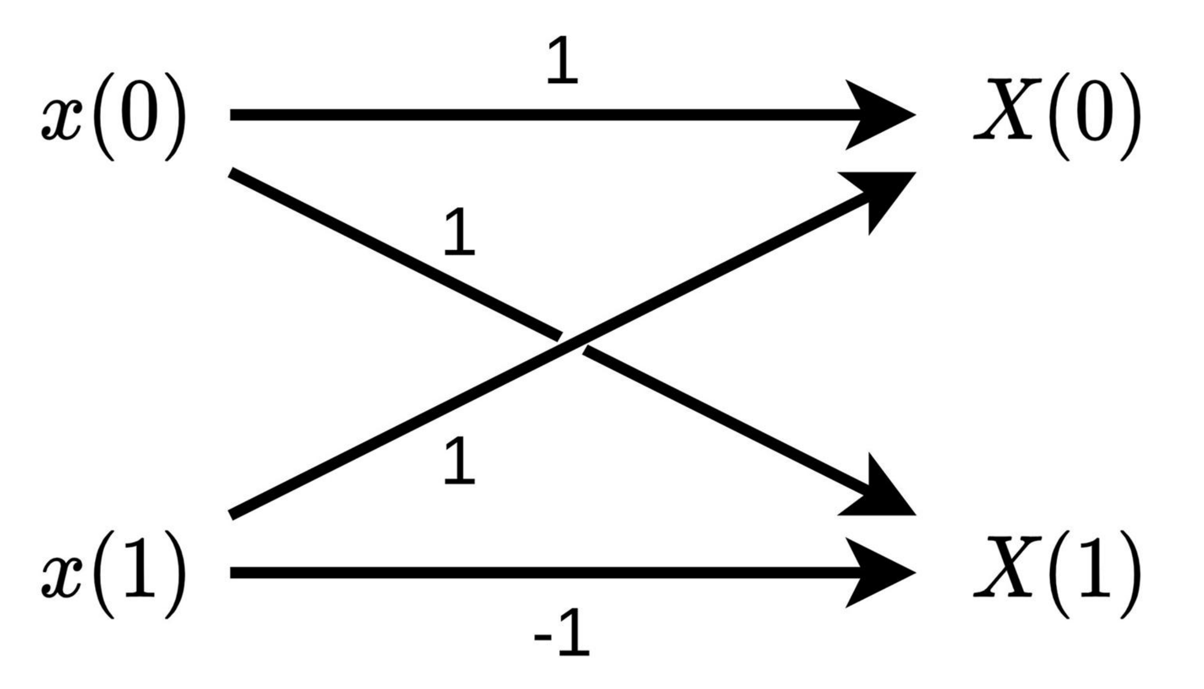
Discrete Fourier Transform

$$X(m) = \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi mn}{N}}$$



Issues with the Discrete Fourier Transform

- Requires N complex multiplications
- Requires N-1 complex additions
- Requires $O(N^2)$ computations

For N= 16 => 160 computations

For N=1024 => 1 048 576 computations

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DFT to FFT

$$\omega_y^x = e^{-\frac{j2\pi x}{y}}$$

$$X(m) = \sum_{n=0}^{N-1} x(n)e^{-\frac{2j\pi mn}{N}} = \sum_{n=0}^{N-1} x(n)\omega_N^{mn}$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(2n)\omega_{N/2}^{2nm} + \sum_{n=0}^{\frac{N}{2}-1} x(2n+1)\omega_{N/2}^{(2n+1)m}$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(2n)\omega_{N/2}^{2nm} + \sum_{n=0}^{\frac{N}{2}-1} x(2n+1)\omega_{N/2}^{2nm}\omega_N^m$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(2n)\omega_{N/2}^{2nm} + \omega_N^m \sum_{n=0}^{\frac{N}{2}-1} x(2n+1)\omega_{N/2}^{2nm}$$

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DFT to FFT

$$X(m) = \sum_{n=0}^{(N/2)-1} x(2n)\omega_{(N/2)}^{mn} + \omega_N^m \sum_{n=0}^{(N/2)-1} x(2n+1)\omega_{(N/2)}^{mn}$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)\omega_{(N/2)}^{mn} - \omega_N^m \sum_{n=0}^{(N/2)-1} x(2n+1)\omega_{(N/2)}^{mn}$$

Requires $O(N^2/2 + N)$ computations

For N= 16 => 144 computations

For N=1024 => 525 312 computations

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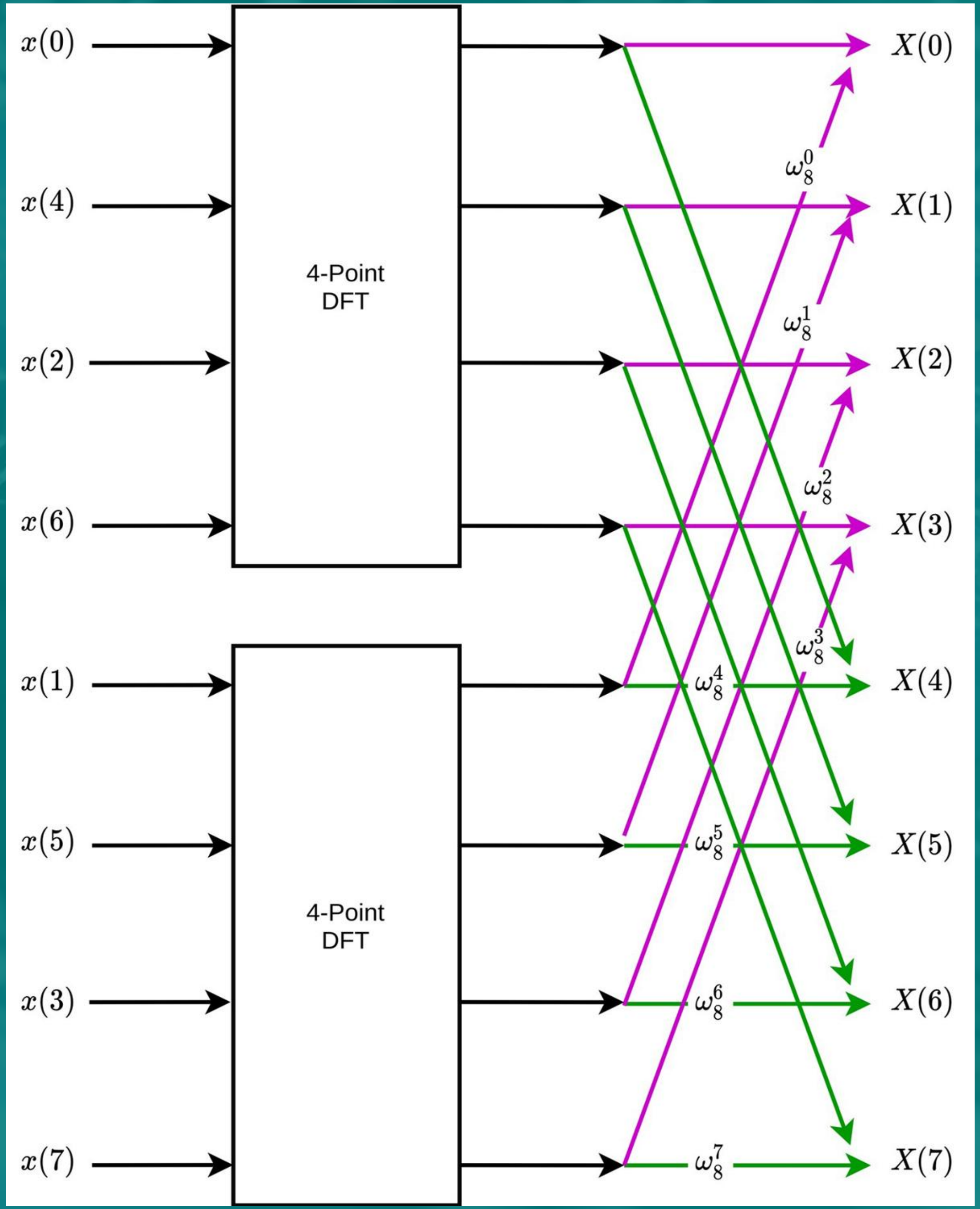
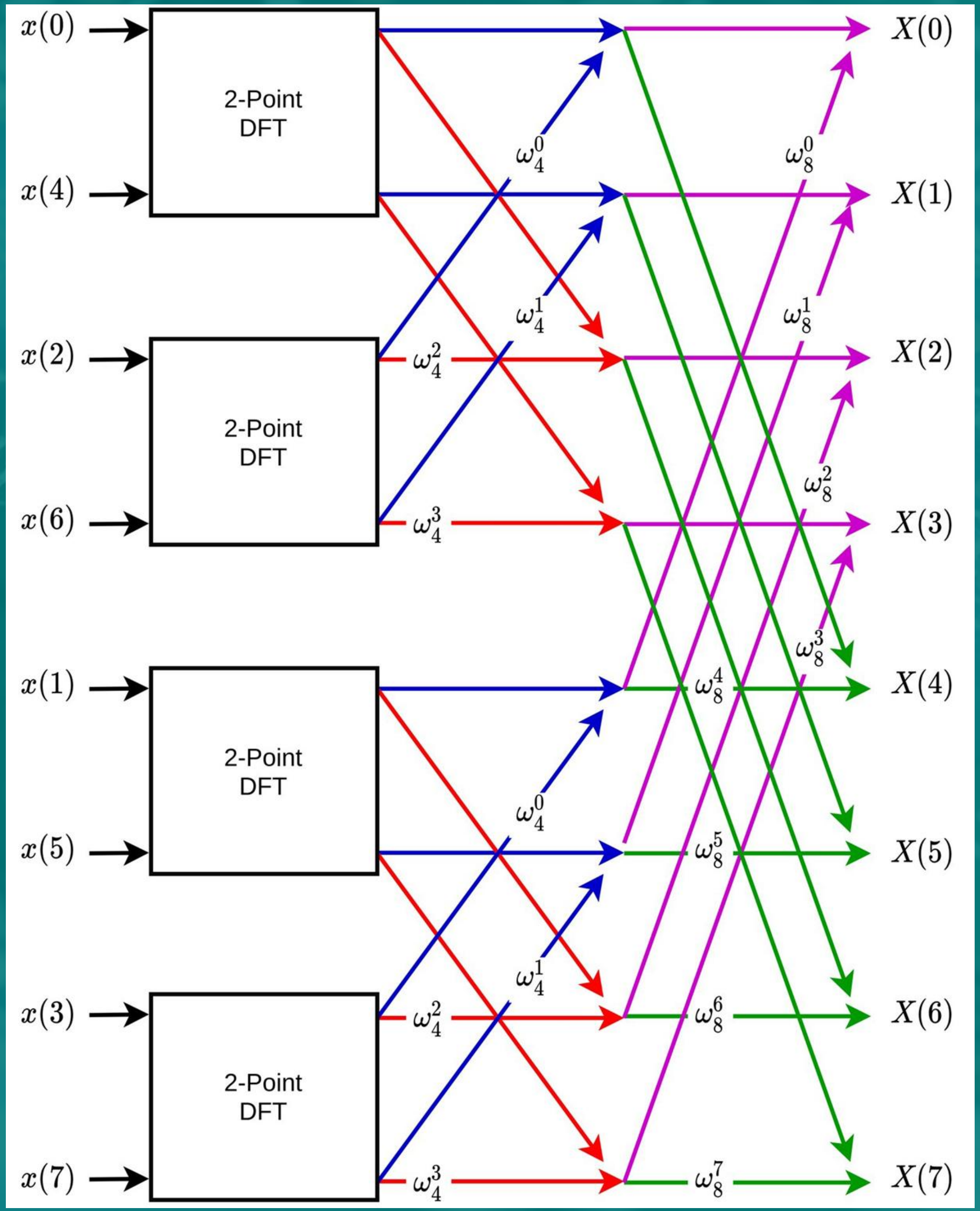
FFT



$x(0)$
$x(4)$
$x(2)$
$x(6)$
$x(1)$
$x(5)$
$x(3)$
$x(7)$
4-Point DFT
4-Point DFT
ω_8^0
ω_8^1
ω_8^2
ω_8^3
ω_8^4
ω_8^5
ω_8^6
ω_8^7
$X(0)$
$X(1)$
$X(2)$
$X(3)$
$X(4)$
$X(5)$
$X(6)$
$X(7)$

FFT



1	2	3	4
$x(0)$	$x(2)$	$x(4)$	$x(6)$

FFT

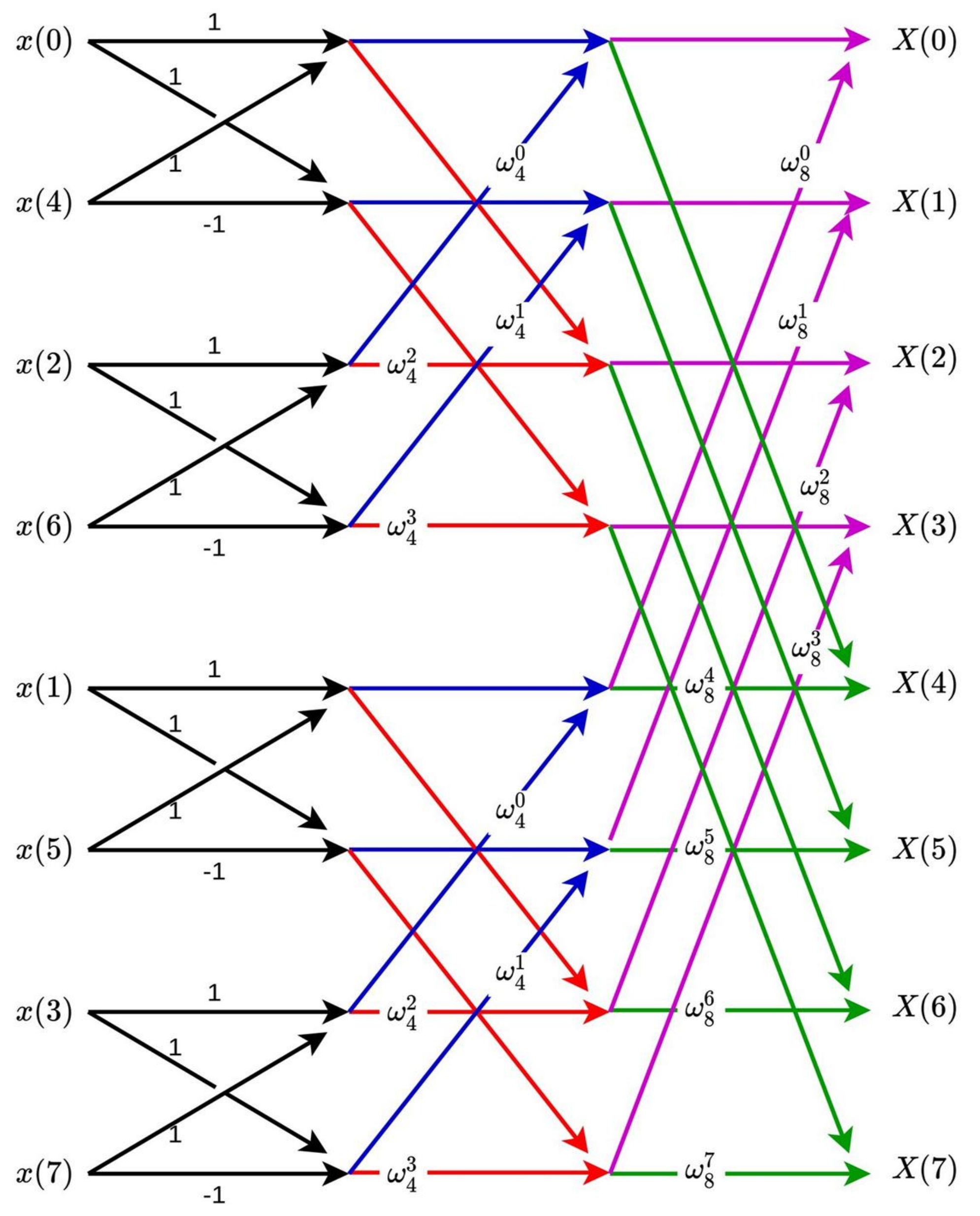
Requires $O(N^2/N + N \log_2(N))$ computations

For large N => $O(N \log_2(N))$ computations

For N= 16 => 64 computations

For N=1024 => 10 240 computations

FFT



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FFT

000	$X(0)$	$x(0)$	000
001	$X(1)$	$x(4)$	100
010	$X(2)$	$x(2)$	010
011	$X(3)$	$x(6)$	110
100	$X(4)$	$x(1)$	001
101	$X(5)$	$x(5)$	101
110	$X(6)$	$x(3)$	011
111	$X(7)$	$x(7)$	111

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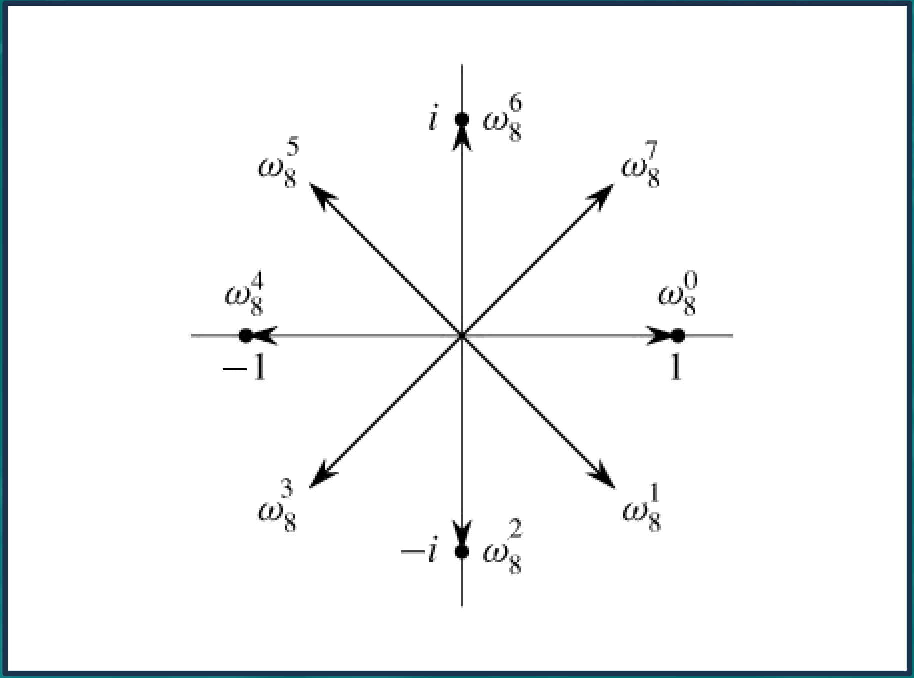
Twiddle Factor

$$\omega_y^x = e^{-\frac{j2\pi x}{y}}$$



Advice

- Design the AAF and scaling circuit, simulate in LTSpice and check specs and operation. Build in hardware and check specs and correct operation.
- Create the FFT algorithm in software (Python) and check correct output with a built-in FFT function. Does your algorithm work correctly (generic i.t.o. N values)
- Figure out how the STM32 works and become a guru.
- Double check each other and integrate in hardware.
- Don’t use TouchGFX => BSP (board support package)
- Pre-report is optional
- Demos + Full report due is on 5th March 2026

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Demo Criteria

Sampling	Correct sampling rate
	No significant aliasing
	Correct voltage input level
	Triple interleaved ADC
FFT	Reasonable spectral profile (sinusoidal frequency sweep)
	Correct frequency of sinusoid confirmed
Display	Basic graph
	Marked intervals on axes

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Report Criteria

<table>
<tr><td rowspan="2">Theoretical background</td><td rowspan="2">Mathematical analysis, Butterfly diagrams, Twiddle factor explanation</td><td>Sampling/Anti-aliasing</td></tr>
<tr><td>FFT</td></tr>
<tr><td rowspan="3">Filter and algorithm implementation</td><td rowspan="3">Filter design. Algorithm described in words, pseudocode, or flow chart</td><td>Anti-aliasing filter</td></tr>
<tr><td>Bit reversal</td></tr>
<tr><td>FFT</td></tr>
<tr><td rowspan="4">Software flow diagrams
+
Code</td><td rowspan="4">Logical (is it well commented?)</td><td>Sampling</td></tr>
<tr><td>Bit reversal</td></tr>
<tr><td>FFT</td></tr>
<tr><td>Display</td></tr>
</table>

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Thank you!

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